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Understanding Petroleum Systems Risk in the Deep Middle Eastern Plays: A Regional Screening Approach

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The Mesozoic and Cenozoic petroleum systems of the Middle East are well renowned for producing large volumes of hydrocarbons to support and supply global demands for oil and gas. Thus far, much of the exploration and production has been focused on these shallower Mesozoic and Cenozoic petroleum systems, helping locate many of the giant and supergiant hydrocarbon fields, particularly along the eastern backbone of the Arabian Peninsula. However, the existing contributions from the Paleozoic are still significant. Vast reserves of gas in the Permo-Triassic Khuff carbonates, for example, are charged from the world-class Silurian Qusaiba Formation which is widespread across Arabia.

Within the public domain, there are large volumes of multi-disciplinary data to assess, understand, analyze, and determine the distribution of petroleum system elements associated with the Mesozoic and Cenozoic plays. The availability of such valuable data begins to fade beyond the Permian, further reducing to isolated ‘pockets’ of data in the Precambrian-Cambrian. This lack of subsurface penetrations into the Paleozoic stratigraphy results in uncertainty in the distribution of petroleum system elements with increasing depth. To date, only clusters of successful Paleozoic discoveries are in Central Arabia (Ghawar area), the Arabian Gulf, Zagros, and Oman.

To further illuminate the exploration potential of this underexplored Paleozoic interval in the Middle East, a plate-wide screening approach has been conducted to help identify ‘zones’ of exploration potential with multiple stacked opportunities for plays. This article will focus on six play fairways which span the Paleozoic and earliest Mesozoic including:

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To further illuminate the exploration potential of this underexplored Paleozoic interval in the Middle East, a plate-wide screening approach has been conducted to help identify ‘zones’ of exploration potential with multiple stacked opportunities for plays. This article will focus on six play fairways which span the Paleozoic and earliest Mesozoic including:

- Middle-Late Ordovician Qasim Formation
- Latest Ordovician (Hirnantian) Sarah Formation
- Early-Middle Devonian Jauf Formation
- Permo-Carboniferous Unayzah Formation
- Permo-Triassic Khuff Formation
- Middle-Late Triassic Jilh Formation

Importance of Locating Gas Plays

Over the next 30 years (2050 and beyond), society will still demand a certain amount of oil and perhaps more importantly, a significant amount of gas as part of the overall energy mix within the energy transition. Crucially, existing gas assets will be insufficient to meet predicted estimates (Figure 1), requiring exploration to focus on locating additional gas reserves in the subsurface. This required supply to meet the future demand is approximately equivalent to 110% of all the gas consumed thus far (Davies and Simmons, 2021). Various demand and supply scenarios are illustrated in Davies and Simmons (2021), with the ‘Paris Mean’ reflecting the mean energy transition scenario across all the demand and supply predictions (Figure 2).

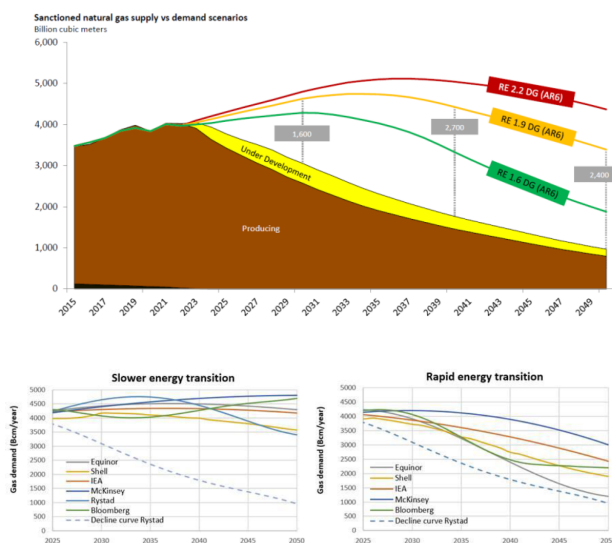


Figure 1—Decline curve of supply from known reserves (Rystad 2P) of gas versus demand curve for gas in a selection of slow energy transition scenarios (top) and a selection of rapid energy transition scenarios (bottom).

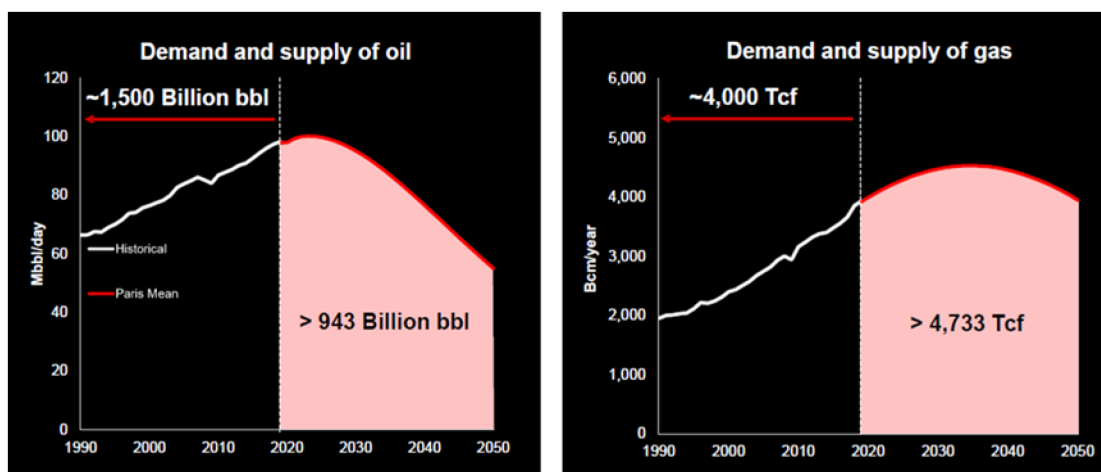


Figure 2—Modified from Davies and Simmons, 2021, highlighting the Paris Mean trajectory for demand and supply of oil and gas up to 2050.

Taking this ‘Paris Mean’ scenario, up to 2050, we will still likely require, more than 940 Bbbl of oil and >4700 Tcf of gas (Figure 2), therefore the oil and gas contribution to the energy mix will play a role for many years to come. Yet where will these required reserves come from? A valuable graph from a global data and analytics platform categorizes the produced and remaining reserves from several highly petroliferous basins across the globe. Within the top five are three Middle Eastern basins which include the Rub al Khali, Widyan, and Zagros basins with significant yet-to-find values. Remaining reserves in these basins are vast with approximately 600 Billion barrels of oil equivalent (Bboe) in the Rub al Khali Basin, ~450 Bboe in the Widyan Basin, and ~150 Bboe in the Zagros Basin (Wood Mackenzie, 2022). This highlights the importance of the Middle East being one of the main contributors to future supply of oil and gas with perhaps a significant proportion of the gas supply being within the Paleozoic stratigraphy. Therefore, the work discussed herein provides a valuable first-pass contribution to help with the effort of locating ‘zones’ of exploration potential.

Paleozoic Petroleum System and Challenges

The Paleozoic stratigraphy of the Middle East represents a significant and underexplored succession. Aside from the Precambrian and early Cambrian stratigraphy which contains petroleum significant carbonates (especially in Oman), much of the stratigraphic succession is dominated by siliciclastics (see Sutcliffe, 2016 and Figure 3) and is related to the gradual movement of Arabia from equatorial latitudes toward southern high latitudes (see Figure 3.3 of Sharland et al., 2001) which induced important glacial periods such as the Late Ordovician (Hirnantian) glaciation.

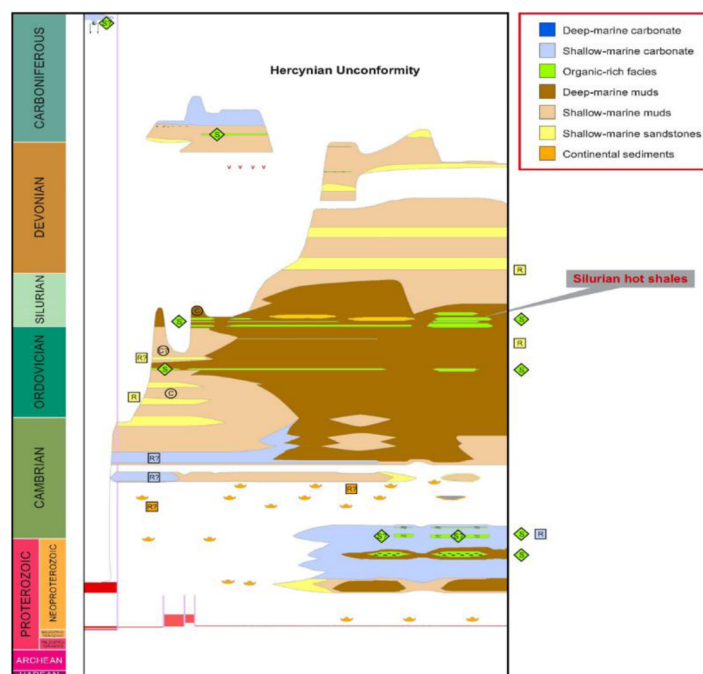


Figure 3—Simplified NW-SE Neftex® chronostratigraphic chart across the Arabian Plate to highlight the dominance of Paleozoic siliciclastic deposition across the Middle East and the impact of differential erosion on the preservation of Paleozoic stratigraphy from the ‘Hercynian’ Unconformity.

The Paleozoic successions contain multiple petroleum systems with some of the most significant located in Oman charged by the Precambrian-Cambrian Huqf Supergroup. Elsewhere across the Arabian Plate, the dominance of gas is derived from the world-class, organic-rich shales of the early Silurian Qusaiba Formation and its equivalents. Its distribution across the Arabian Peninsula relates in part to its preservation beneath the Late Paleozoic ‘Hercynian’ unconformity, the occurrence of major structural depressions and paleobathymetric variations related to the Late Ordovician (Hirnantian) glaciation. Depending on its location in the subsurface, it can charge both younger and older Paleozoic stratigraphy (see [Sutcliffe and Cousins, 2018](#)). At present-day depths, large areas of the Silurian are overmature and therefore require an understanding of charge timing, the nature of migration pathways either laterally, vertically or multiple ‘mobile’ phases of charge related to structural development of the region.

Subsurface Expression of the Paleozoic Stratigraphy

Across the Middle East, the expression and preservation of the Paleozoic stratigraphy is varied, which is related to the impact of the ‘Hercynian’ orogeny (see [Sutcliffe and Cousins, 2018](#)). Additionally, the Paleozoic stratigraphy is impacted by later significant events such as Permo-Triassic rifting in the Palmyrides. These key events in Arabian Plate history have led to a geologically complex Paleozoic succession beneath the ‘Hercynian’ unconformity, resulting in multiple episodes of basin lineament reactivation.

The ‘Hercynian’ Orogeny, which is the result of the collision of Gondwana and Laurussia, formed the Pangea supercontinent. This collision resulted in the uplift of Paleozoic stratigraphy, non-deposition, and differential erosion across the plate. As expressed in [Figure 3](#), in some places, the ‘Hercynian’ unconformity has completely eroded many of the Paleozoic petroleum system elements, incising down into the underlying Cambrian stratigraphy. Therefore, the expression of the preserved stratigraphy beneath the ‘Hercynian’ unconformity, i.e., the subcrop, must be considered in any plate-wide screening effort to determine exploration potential in places where petroleum system elements may have been eroded.

The Plate-wide Screening Approach

In the identified six fairways spanning the Paleozoic to earliest Mesozoic, a plate-wide screening methodology is applied, in which the highest petroleum system element risk is carried forward through to the final Combined Common Chance Map (CCCM).

This approach illuminates ‘zones’ across Arabia where an individual play has maximum potential. Each of the input parameters e.g., source rock presence, is derived from the integration of multi-disciplinary subsurface datasets that are tied to a stratigraphic framework. Each of the identified play fairways were assessed on an individual basis with a focus on charge from the Silurian Qusaiba Formation and its equivalents, with an arbitrary migration of 80km from the source kitchen. The results from this individual play fairway analysis approach were then stacked to allow the final output to be engineered, which highlighted areas of stacked play potential (Figure 4). Within the results and conclusion section, the outcomes of this process and associated petroleum systems risk for each fairway will be discussed in more detail.

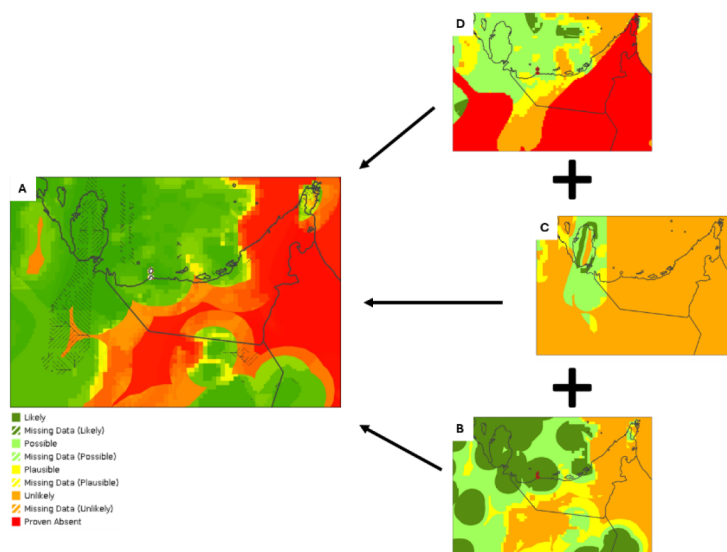


Figure 4—A worked example of the individual final CCCM for the fairways (B, C, and D), these are then stacked on top of one another to illuminate ‘zones’ of stacked play potential (A).

Paleozoic-earliest Mesozoic Fairway Results

By reviewing the final CCCM of each fairway, ‘zones’ of potential can be quickly located across the Arabian Plate (Figure 5). These ‘zones’ can then undergo more detailed investigation utilizing proprietary datasets to identify exploration potential, undergo risk analysis, and develop a portfolio of opportunity.

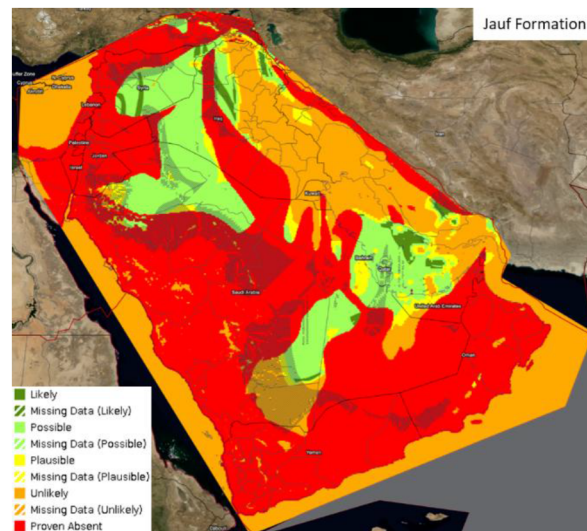


Figure 5—A simplified worked example of the Final Combined Common Chance maps for the Early-Middle Devonian Jauf Formation. Note, lateral and up-dip migration from the Qusaiba Formation source kitchens to a distance of 80km indicated by the black lines.

Middle-Late Ordovician Qasim Formation

The Middle-Late Ordovician Qasim Formation and its equivalents represent a set of alternating sandstone and shales deposited on a stable shelf setting that is well preserved in the subsurface across central and northern Arabia ([Figure 3](#)). Preserved equivalents of the Qasim Formation in Oman are largely restricted to the axis of the major salt basins due to erosion from the late Carboniferous ‘Hercynian’ unconformity. Plate-wide, there is considerable potential represented in the Qasim fairway, with widespread potential across Saudi Arabia, Bahrain, Qatar, western and northern Iraq, Syria, and southern Turkey. However, as this screening analysis considers charge from the overlying Silurian Qusaiba Formation, areas where the Qasim Formation and its equivalents are preserved on major paleo-highs could be more favorable, where charge from the onlapping Silurian in the adjacent topographic lows can migrate up-dip and along major basement faults. Other favourable ‘zones’ for the Qasim are areas that have undergone rifting such as the Palmyrides which may favor the juxtaposition of source rock and reservoir as well as generating potential trapping mechanisms. Other considerations for charge may include downward-migration where pressure differences exist between the overlying source rock (being over pressured) and atmospheric pressure existing in the underlying reservoir (Yangfang et al., 2009). An effective seal is not a risk for this play due to the alternation of sandstones and thick shale sequences ([Figure 3](#)), however, in northern Iraq ([Omer and Friis, 2014](#)) and the Rub al Khali Basin ([Hui et al., 2016](#)), quartz cementation represents a major risk for this play due to depth of burial reducing reservoir quality, requiring chlorite coating of sand grains or early charge to preserve primary porosity.

Latest Ordovician (Hirnantian) Sarah Formation

Potential in the Latest Ordovician (Hirnantian) Sarah Formation, compared to the Qasim Formation, is limited to northern Saudi Arabia and western Iraq. As this reservoir is glacially influenced, related to the movement of Arabia to southern high latitudes ([Figure 3.3 of Sharland et al., 2001](#)), it is vitally important to understand glacial maxima and minima as this will have important implications on the sedimentological model to predict the distribution and presence of potential reservoir facies (e.g. [Huuse et al., 2012](#); [Kurjanski et al., 2020](#)). Additionally, the glaciers tended to erode deeply into very mud-prone underlying stratigraphy ([Figure 3](#)) and it is likely that the facies associated with the Sarah Formation will be very heterogeneous and therefore pose limited connectivity between reservoir bodies. As discussed above in the Qasim Formation,

charge (whether up-dip, downward migration or juxtaposition of source rock) will be the key risk to play potential.

Early-Middle Devonian Jauf Formation

Moving through the stratigraphic succession to the Early-Middle Devonian Jauf Formation, much of the Devonian stratigraphy has been eroded (Figure 3) due to the impact of the ‘Hercynian’ unconformity and the earlier Mid-Devonian unconformity across Saudi Arabia, related to plate reorganization associated with subduction and back-arc rifting of the Proto-Tethys (Sharland et al., 2001). This uplift affected the long-lived Ghawar High, causing it to be exposed and eroded (Al-Husseini, 2000; Sharland et al., 2001). The Early and Middle Devonian stratigraphy was eroded during this event, particularly in southern Saudi Arabia, but the amount of stratigraphy removed diminishes towards the east. The presence of reservoir is a major risk due to the complex geology and preservation of the Jauf Formation in the subcrop beneath the ‘Hercynian’ unconformity (Figure 5). Notably, the Devonian stratigraphic package is absent across many of the northward paleo-topographic highs across Arabia and is only preserved in topographic lows and the flanks of structures (Figure 5).

Permo-Carboniferous Unayzah Formation

The deposits of the Permo-Carboniferous Unayzah Formation and its equivalents vary from glacially influenced like that of the latest Ordovician (Hirnantian) Sarah Formation to fluvial and/or aeolian and overly the ‘Hercynian’ unconformity. The Unayzah formation is conventionally split into three members (Unayzah A, B and C), but a fourth between the A and B members was identified by Melvin and Sprague (2006) named the Unnamed Middle Unayzah Member. The oldest member is assigned to the Unayzah C, consisting of a monotonous succession of quartzose sandstones deposited in a glacio-fluvial braided plain. The overlying Unayzah B Member contains a diverse and complex suite of depositional elements ranging from terminal moraines, glacially-influenced lacustrine deposits, and diamictites with less common braided stream and aeolian deposits (Melvin and Sprague, 2006). The Unnamed Middle Unayzah Member consists of very shallow lacustrine silts and sandstones deposited in an extensive alluvial braided plain with the Unayzah A Member indicative of aeolian sandstones (Macdonald et al, 2010) with braided fluvial deposits also recorded. This highlights the complexity in the Unayzah Formation reservoir and therefore, requires a strong understanding of depositional and sedimentological models to identify where the best reservoir packages may be in the subsurface.

Unlike the Sarah Formation, the ‘zones’ for exploration potential across Arabia associated with the Unayzah are vast in comparison. In Omani equivalent stratigraphy, additional risks associated with this play are associated with biodegradation and the formation of heavy oil due to shallow burial and the infiltration of meteoric waters. It is important to note that in Oman, almost all the Silurian stratigraphy has been eroded and therefore, the oil will be associated with charge from the Precambrian-Cambrian Huqf Supergroup.

The Khuff and Jilh formations

Moving into the early Mesozoic signals a dramatic change in facies from the dominant siliciclastic deposits that comprise the Paleozoic to shallow-marine carbonates. This is related to the gradual movement of the Arabian Plate northward to equatorial latitudes and favorable climatic conditions for carbonate deposition (Figure 3.3 of Sharland et al., 2001). The Khuff Formation represents broad carbonate deposition over a low-angle ramp associated with a vast epeiric sea across an area of 3.7 million km². The low-angled dip of this carbonate platform made the Arabian Plate extremely susceptible to eustatic sea-level change resulting in cyclic carbonate deposition. Whilst the presence of reservoir is not a major risk for this fairway, it is important to determine the distribution of grainy/oolitic facies alongside the degree of karstification related to exposure of the carbonate platform during periods of lower sea-level for enhanced reservoir quality characteristics. Within the public domain, there is very little data constraint on the presence of grainy/oolitic

facies. This highlights the need to integrate datasets and methodologies such as hydrocarbon occurrences, tectonic lineaments, and sequence stratigraphy with a holistic Earth System Science approach (see [Jennings, 2021](#)) to understand tidal and bed shear stresses to predict the likelihood of grainy/oolitic facies. A major risk to the carbonates of the Khuff Formation and its equivalents is gas souring. Within the Khuff Formation, there are multiple significant anhydrite beds that, at subsurface temperatures of >130–140°C and in the presence of methane, have undergone thermochemical sulfate reduction (TSR), which has resulted in H₂S concentrations of up to 98% (e.g. [Worden and Smalley, 1996](#)), destroying hydrocarbon accumulations. Therefore, understanding the distribution of anhydrite, migration pathways and the subsurface thermal regime are important parameters to determine in the exploration workflow. Similar risks (reservoir quality and gas souring) are associated with the Jilh Formation, equivalent to the Kurrachine and Geli Khani formations, with other petroleum system risks related to the patchy nature of the seal for the Jilh carbonate reservoir. Seal lithology and age varies across the plate from interbedded transgressive and highstand shales, to evaporites in the Palmyrides and Early Jurassic aged evaporites through Iraq.

Considerations

Whilst in each fairway ‘zones’ of maximum potential have been identified, it is important to highlight that this first-pass, plate-wide screening approach has focused on charge from the Silurian Qusaiba Formation and its equivalents. Aside from the Silurian, there are multiple source rock horizons that have the potential to charge these deep gas plays. Whilst proven in Oman, could equivalents to the Huqf Supergroup source rocks be preserved elsewhere across Arabia? Available seismic from [Stewart et al. \(2016\)](#), interprets a deep Precambrian rift structure within the Rub al Khali Basin, whilst wells have penetrated Precambrian sections in Yemen e.g. Qinab-1 ([Petroleum Exploration and Production Authority, 2013](#)), Jordan e.g. Wadi Sirhan-3 ([Andrews et al., 1991](#); [Al-Husseini, 2010](#); and [Luning et al. 2005](#)) as well as in Israel, the Sinai Peninsula and southern Turkey. However, in Jordan, the Precambrian-Cambrian succession is dominated by coarse siliciclastics and is likely diluting any organic enrichment. Several authors e.g., [Mahmoud et al. \(1992\)](#) and [Al Ghamdi et al. \(2024\)](#) have investigated the potential of the Ordovician Hanadir Member and Ra'an Member shales. Reported Total Organic Carbon (TOC) values in these members are low (<1 wt.%) and have not been directly linked to charging hydrocarbon accumulations across Arabia. In isolated areas, could these members have elevated TOC and represent source rock potential? Zooming out of the Arabian Plate, the Frasnian Awaynat Wanin Formation is a known source rock in North Africa with up to 11% TOC, could similar values be present in the ‘pre’-Mesopotamian basin with speculative organic enrichment proposed in the Abu Safah-29 well ([Al-Hajri and Owens, 2000](#)). Finally, isolated ‘pods’ of organic enrichment ([Aqrawi et al., 2010](#)) in the Permo-Triassic carbonate platform related to differential bathymetry on a low-angle carbonate ramp has been speculated to generate organic-rich facies, whilst in the Zagros, gas and condensate accumulations in the Dalan and Kangan formations have been tied to source rock facies within the Kangan Formation ([Ahanjan et al., 2017](#)).

Migration from the source kitchen in the screening outputs are based on lateral migration and does not account for vertical migration or the utilization of faults to juxtapose stratigraphically older reservoir in proximity with stratigraphically younger source rock to act as migration pathways. This juxtaposition via faults for older reservoir and younger source rock is particularly important for the Middle-Late Ordovician Qasim Formation. First-pass assessments of this geometry can be assessed using the NefteX Predictions portfolio by integrating depth structure grids and tectonic elements within geospatial software. By removing or buffering the distance between the faults, the depth surfaces can be ‘re-stitched’ to illuminate zones where the younger source rock can be juxtaposed next to, or near, stratigraphically older reservoir and fluids can migrate.

Maturity of the Silurian is based on present-day depth and therefore the source kitchens are extensively overmature. A more thorough investigation in timing of charge, migration pathways, and the potential

for multiple ‘mobile’ phases of charge related to structural development of the region is an important consideration.

Finally, understanding differential topography across the region to locate areas where the Silurian is onlapping topographic highs and therefore, the potential to charge reservoirs up-dip whilst combining this with a strong understanding of the preserved subcrop stratigraphy is key to help further delineate play potential.

Conclusions

Understanding deep gas systems within the Middle East requires a regional approach to fully appreciate and contextualize the distribution of petroleum system elements and processes. We can use a play screening workflow to not only assess the likely presence and effectiveness of each petroleum system element but understand what the key risks for each area will be and how this will influence each petroleum systems development. This methodology has identified multiple ‘zones’ across the Arabian Plate where deep gas plays could be viable provided a sufficient trapping mechanism is in place. By stacking each of the individual fairways (Figure 6), we can high-grade areas across the Arabian Peninsula which may have multiple, stacked play potential areas in which proprietary datasets can then be used to test and validate identified ‘zones’. This approach has identified stacked play potential on and in proximity to basin highs such as the Qatar Arch and multiple play potential in rifted basin margins such as the Palmyrides (Figure 6). Areas across the Arabian Plate which have limited stacked play potential is a result of erosion from the ‘Hercynian’ Unconformity (removal and/or non-deposition of stratigraphy) and where reservoir effectiveness and charge pose a significant risk due to depth of burial (Figure 6).

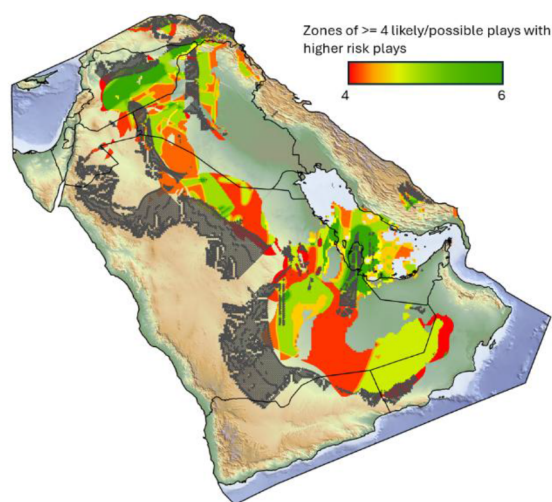


Figure 6—Showing areas across the Arabian Plate where ‘zones’ of more than 4 plays exist to help illuminate areas where multiple play potential may exist.

This approach has identified that the Paleozoic stratigraphy of the Middle East has potential plate-wide and perhaps the ability to replicate the exploration and production success of the Mesozoic and Cenozoic intervals.

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